Threat Modeling Report

SecureShare - CYC386 Assignment

Project: SecureShare
Version: 1.0
Date: October 29, 2026
Course: CYC386 Secure Software Design and Development
Methodology: STRIDE + Attack Trees + CVSS v3.1

Table of Contents

1. System Overview
2. Context-Level Data Flow Diagram
3. Level-1 Data Flow Diagram
4. STRIDE Analysis
5. Attack Trees
6. CVSS v3.1 Scoring
7. Risk Register
8. Residual Risk Analysis
9. Threat to PNE Mapping
10. Mandatory Vulnerability Classes

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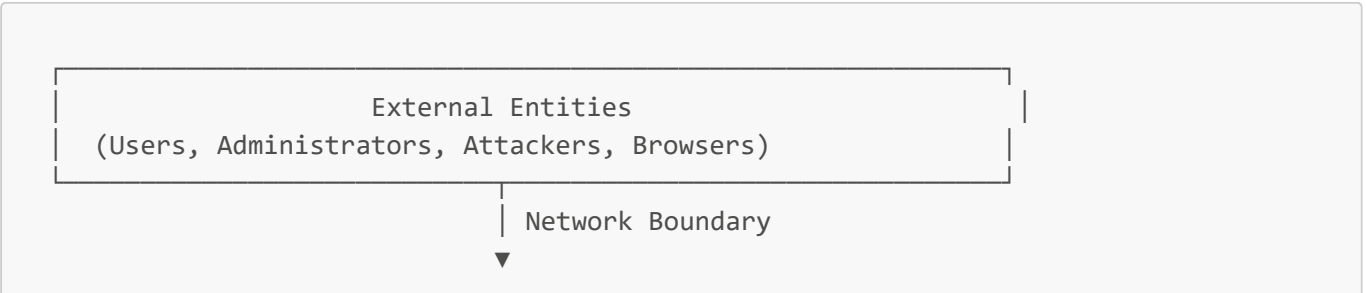
1. System Overview

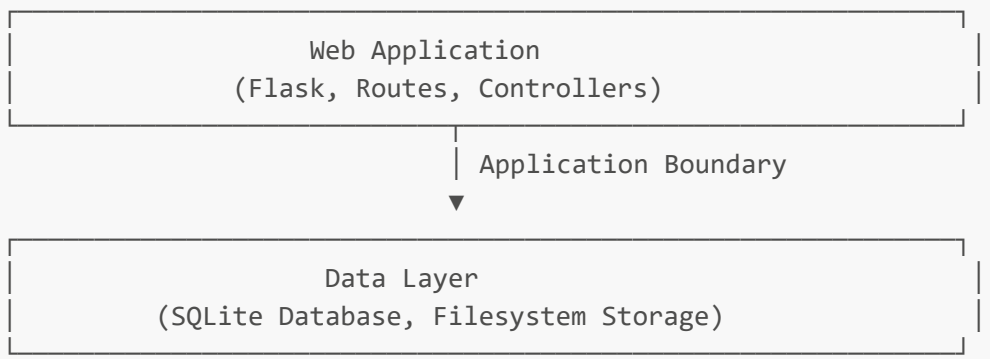
1.1 System Description

SecureShare is a Flask-based web application for secure file sharing with the following components:

- **Web Server:** Flask development server (127.0.0.1:5000)
- **Application Layer:** Python Flask with blueprints (auth, main, admin)
- **Database:** SQLite (instance/secureshare.db)
- **File Storage:** Local filesystem (instance/uploads/)
- **Authentication:** Flask-Login with PBKDF2 password hashing
- **Session Management:** HTTP-only, SameSite cookies
- **CSRF Protection:** Flask-WTF CSRF tokens

1.2 Trust Boundaries



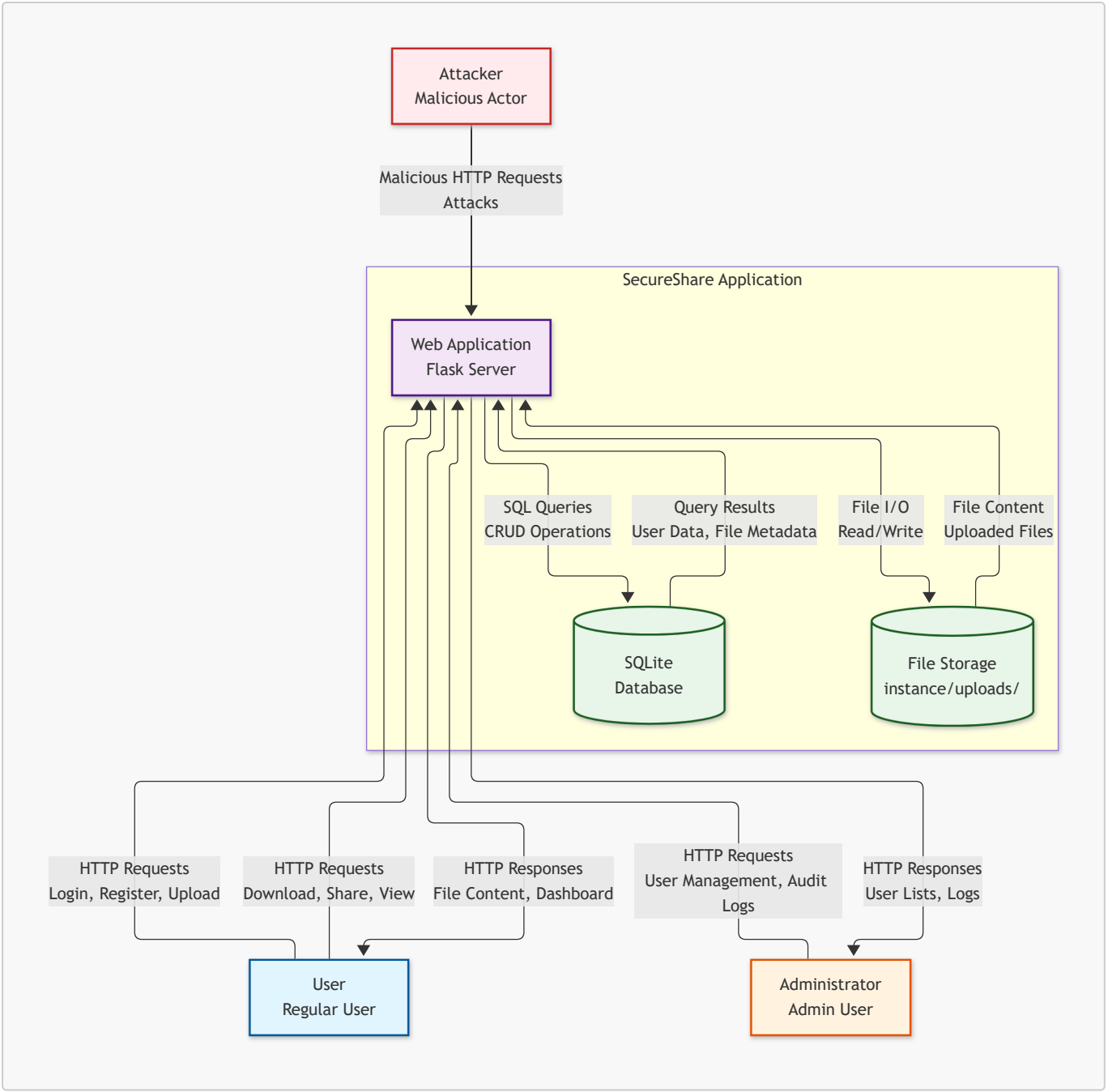


1.3 Data Elements

- **User Credentials:** Username, password (hashed), email
- **User Sessions:** Session tokens, cookies
- **File Content:** Uploaded files (text, documents, images)
- **File Metadata:** Filename, size, content type, description
- **Audit Logs:** User actions, timestamps, IP addresses, user agents
- **Sharing Permissions:** Read/write access controls

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2. Context-Level Data Flow Diagram



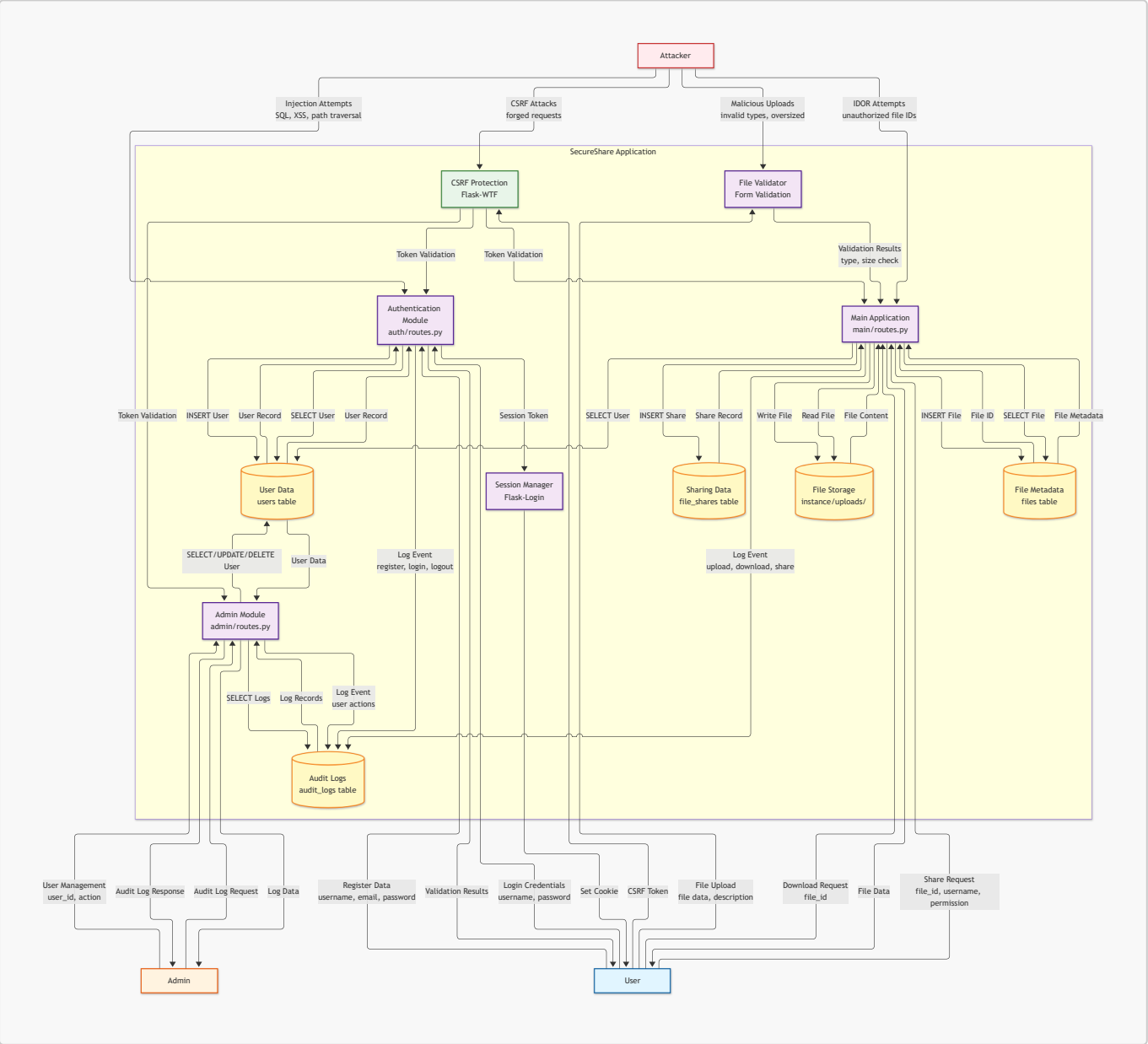
2.1 Context DFD Elements

Element	Type	Description
User	External Entity	Regular user accessing file sharing features
Admin	External Entity	Administrator managing users and viewing audit logs
Attacker	External Entity	Malicious actor attempting to compromise the system
Web Application	Process	Flask web server handling HTTP requests
SQLite Database	Data Store	Database storing users, files, shares, audit logs
File Storage	Data Store	Filesystem storing uploaded files

2.2 Data Flows

Flow	Source	Destination	Data
F1	User	Web Application	Login credentials, registration data, file uploads
F2	Web Application	User	Dashboard, file lists, download responses
F3	Admin	Web Application	User management commands, audit log requests
F4	Web Application	Admin	User lists, audit log data
F5	Attacker	Web Application	Malicious payloads, attack vectors
F6	Web Application	Database	SQL queries (INSERT, SELECT, UPDATE, DELETE)
F7	Database	Web Application	Query results, user data, file metadata
F8	Web Application	File Storage	File write operations (uploads)
F9	File Storage	Web Application	File read operations (downloads)

3. Level-1 Data Flow Diagram



3.1 Level-1 DFD Elements

Element	Type	Description	Code Reference
Authentication Module	Process	Handles user registration, login, logout	<ref_file file="D:\SSD LAB MIDTERM\app\auth\routes.py" />
Main Application	Process	Handles file upload, download, sharing	<ref_file file="D:\SSD LAB MIDTERM\app\main\routes.py" />
Admin Module	Process	Handles user management, audit logs	<ref_file file="D:\SSD LAB MIDTERM\app\admin\routes.py" />
File Validator	Process	Validates file type, size, filename	<ref_file file="D:\SSD LAB MIDTERM\app\main\routes.py" lines="12-26" />
Session Manager	Process	Manages user sessions via Flask-Login	<ref_file file="D:\SSD LAB MIDTERM\app_*init*_.py" lines="26-34" />
CSRF Protection	Process	Validates CSRF tokens on forms	<ref_file file="D:\SSD LAB MIDTERM\app_*init*_.py" line="24" />
User Data	Data Store	User accounts and credentials	<ref_file file="D:\SSD LAB MIDTERM\app\models.py" lines="9-39" />
File Metadata	Data Store	File information and ownership	<ref_file file="D:\SSD LAB MIDTERM\app\models.py" lines="42-77" />
Sharing Data	Data Store	File sharing permissions	<ref_file file="D:\SSD LAB MIDTERM\app\models.py" lines="80-99" />
Audit Logs	Data Store	Security event logs	<ref_file file="D:\SSD LAB MIDTERM\app\models.py" lines="102-117" />
File Storage	Data Store	Uploaded file content	<ref_file file="D:\SSD LAB MIDTERM\app\config.py" line="21" />

4. STRIDE Analysis

4.1 External Entities

User

Threat	STRIDE Category	Description	Likelihood	Impact	Risk
Credential Theft	S	Attacker steals user credentials via phishing, keylogging	Medium	High	**High**

Threat	STRIDE Category	Description	Likelihood	Impact	Risk
Session Hijacking	S	Attacker compromises user session cookie	Medium	High	**High**
CSRF Attacks	S	Attacker tricks user into unintended actions	Medium	Medium	Medium
Weak Passwords	S	User chooses weak, guessable password	High	Medium	Medium
Malicious File Upload	T	User uploads malicious file (virus, malware)	Low	Medium	Low

Administrator

Threat	STRIDE Category	Description	Likelihood	Impact	Risk
Privilege Escalation	E	Attacker gains admin privileges	Low	Critical	**High**
Admin Credential Compromise	S	Attacker steals admin credentials	Low	Critical	**High**
Abuse of Admin Privileges	I	Admin intentionally misuses privileges	Low	High	Medium
Audit Log Tampering	T	Admin modifies or deletes audit logs	Low	High	Medium

Attacker

Threat	STRIDE Category	Description	Likelihood	Impact	Risk
Brute Force Attack	S	Automated password guessing	High	Medium	**High**
SQL Injection	I	Injecting malicious SQL queries	Low	Critical	**High**
XSS Attack	I	Injecting malicious scripts	Medium	High	**High**
CSRF Attack	S	Forging authenticated requests	Medium	Medium	Medium
Path Traversal	I	Accessing files outside intended directory	Low	High	Medium
IDOR	I	Accessing unauthorized resources via IDs	Medium	High	**High**

4.2 Processes

Authentication Module

Threat	STRIDE Category	Description	Likelihood	Impact	Risk	Mitigation Status
Password Hashing Weakness	I	Weak password hashing algorithm	Low	Critical	**High**	✅ Mitigated (PBKDF2-SHA256)
SQL Injection in Auth	I	SQL injection in login/register	Low	Critical	**High**	✅ Mitigated (SQLAlchemy ORM)
Session Fixation	S	Attacker sets known session token	Low	High	Medium	✅ Mitigated (Flask-Login)
Missing Rate Limiting	D	Brute force attacks on login	High	Medium	**High**	⚠️ Partial (logging only)
Credential Exposure in Logs	I	Passwords logged in plain text	Low	Critical	**High**	✅ Mitigated (hashed only)
Registration Bypass	T	Skip registration validation	Low	High	Medium	✅ Mitigated (form validation)

Main Application (File Operations)

Threat	STRIDE Category	Description	Likelihood	Impact	Risk	Mitigation Status
IDOR in File Access	I	Access unauthorized files via ID manipulation	Medium	High	**High**	✅ Mitigated (access control checks)
File Type Bypass	I	Upload executable files as allowed types	Medium	High	**High**	✅ Mitigated (whitelist validation)
Path Traversal in Upload	I	Upload files to arbitrary locations	Low	Critical	**High**	✅ Mitigated (secure_filename + random token)
File Size DoS	D	Upload large files to exhaust storage	Medium	Medium	Medium	✅ Mitigated (16MB limit)

Threat	STRIDE Category	Description	Likelihood	Impact	Risk	Mitigation Status
Malicious File Upload	T	Upload malware or malicious content	Low	High	Medium	⚠️ Partial (type check only)
Unrestricted Download	I	Download files without authorization	Medium	High	**High**	✅ Mitigated (access control)
File Metadata Exposure	I	Expose sensitive file metadata	Low	Medium	Low	✅ Mitigated (authorization checks)

Admin Module

Threat	STRIDE Category	Description	Likelihood	Impact	Risk	Mitigation Status
Privilege Escalation	E	Non-admin accesses admin functions	Low	Critical	**High**	✅ Mitigated (@admin_required)
Admin Self-Deletion	I	Admin deletes own account causing lockout	Low	High	Medium	✅ Mitigated (self-check)
Admin Self-Demotion	I	Admin removes own admin privileges	Low	High	Medium	✅ Mitigated (self-check)
Unauthorized User Deletion	I	Delete users without proper authorization	Low	High	Medium	✅ Mitigated (role check)
Audit Log Tampering	T	Modify or delete audit logs	Low	High	Medium	⚠️ Partial (DB integrity only)

File Validator

Threat	STRIDE Category	Description	Likelihood	Impact	Risk	Mitigation Status
MIME Type Spoofing	I	File pretends to be allowed type	Medium	High	**High**	⚠️ Partial (extension check only)

Threat	STRIDE Category	Description	Likelihood	Impact	Risk	Mitigation Status
File Size Bypass	I	Exceed size limit via manipulation	Low	Medium	Medium	✅ Mitigated (Werkzeug enforcement)
Filename Injection	I	Malicious filenames containing path traversal	Low	High	Medium	✅ Mitigated (secure_filename)
Double Extension Attack	I	file.exe.txt bypasses validation	Medium	Medium	Medium	⚠️ Partial (extension check)

Session Manager

Threat	STRIDE Category	Description	Likelihood	Impact	Risk	Mitigation Status
Session Hijacking	S	Steal session cookie via XSS/network	Medium	High	**High**	✅ Mitigated (HTTP-only)
Session Fixation	S	Attacker forces known session ID	Low	High	Medium	✅ Mitigated (Flask-Login)
Session Timeout Bypass	S	Extend session beyond timeout	Low	Medium	Medium	✅ Mitigated (1-hour timeout)
Weak Session Keys	I	Predictable session tokens	Low	Critical	**High**	✅ Mitigated (secure random)

CSRF Protection

Threat	STRIDE Category	Description	Likelihood	Impact	Risk	Mitigation Status
CSRF Token Bypass	S	Bypass CSRF validation	Low	High	Medium	✅ Mitigated (Flask-WTF)
Token Prediction	I	Guess CSRF tokens	Low	High	Medium	✅ Mitigated (cryptographic random)
Token Exposure	I	CSRF token leaked via XSS	Medium	Medium	Medium	⚠️ Partial (no token-specific protection)

4.3 Data Stores

SQLite Database

Threat	STRIDE Category	Description	Likelihood	Impact	Risk	Mitigation Status
SQL Injection	I	Inject malicious SQL queries	Low	Critical	**High**	✅ Mitigated (SQLAlchemy ORM)
Database File Access	I	Direct access to database file	Low	Critical	**High**	⚠️ Partial (filesystem permissions)
Database Corruption	D	Database becomes corrupted	Low	High	Medium	⚠️ Partial (no backups)
Unauthorized Read	I	Read database without authorization	Low	Critical	**High**	⚠️ Partial (filesystem permissions)
Data Exfiltration	I	Extract sensitive data from database	Low	Critical	**High**	⚠️ Partial (filesystem permissions)

File Storage (instance/uploads/)

Threat	STRIDE Category	Description	Likelihood	Impact	Risk	Mitigation Status
Direct File Access	I	Access files via direct URL	Low	High	Medium	✅ Mitigated (outside web root)
Path Traversal	I	Access files outside upload directory	Low	Critical	**High**	✅ Mitigated (secure_filename)
File Deletion	D	Delete files from filesystem	Low	Medium	Medium	⚠️ Partial (filesystem permissions)
Storage Exhaustion	D	Fill disk with uploads	Medium	Medium	Medium	✅ Mitigated (size limits)
Malicious File Execution	T	Execute uploaded file	Low	Critical	**High**	✅ Mitigated (outside web root)

Audit Logs

Threat	STRIDE Category	Description	Likelihood	Impact	Risk	Mitigation Status
Log Tampering	T	Modify or delete audit logs	Low	High	Medium	⚠️ Partial (DB integrity only)
Log Injection	I	Inject fake log entries	Low	Medium	Medium	⚠️ Partial (no integrity checks)
Log Deletion	D	Delete audit logs	Low	High	Medium	⚠️ Partial (no additional protection)
Sensitive Data in Logs	I	Log sensitive information (passwords)	Low	Critical	**High**	✅ Mitigated (passwords not logged)

4.4 Data Flows

HTTP Requests/Responses

Threat	STRIDE Category	Description	Likelihood	Impact	Risk	Mitigation Status
Man-in-the-Middle	I	Intercept and modify HTTP traffic	Medium	High	**High**	⚠️ Partial (HTTPS not enforced in dev)
Request Tampering	T	Modify HTTP requests in transit	Medium	High	**High**	⚠️ Partial (HTTPS optional)
Response Tampering	T	Modify HTTP responses in transit	Medium	High	**High**	⚠️ Partial (HTTPS optional)
Eavesdropping	I	Intercept sensitive data in transit	Medium	High	**High**	⚠️ Partial (HTTPS optional)
Replay Attacks	R	Replay valid requests	Low	Medium	Medium	⚠️ Partial (no nonce/timestamp)

SQL Queries

Threat	STRIDE Category	Description	Likelihood	Impact	Risk	Mitigation Status
SQL Injection	I	Inject malicious SQL via query parameters	Low	Critical	**High**	✅ Mitigated (SQLAlchemy ORM)

Threat	STRIDE Category	Description	Likelihood	Impact	Risk	Mitigation Status
Database Connection Theft	S	Steal database connection credentials	Low	Critical	**High**	✅ Mitigated (environment variables)

File I/O Operations

Threat	STRIDE Category	Description	Likelihood	Impact	Risk	Mitigation Status
Path Traversal	I	Access files outside intended directory	Low	Critical	**High**	✅ Mitigated (secure_filename)
Race Conditions	R	TOCTOU race conditions in file operations	Low	Medium	Low	⚠️ Partial (no specific protection)
Symbolic Link Attacks	I	Use symlinks to access unauthorized files	Low	High	Medium	⚠️ Partial (no symlink checks)

4.5 Trust Boundaries

Network Boundary (External → Web Application)

Threat	STRIDE Category	Description	Likelihood	Impact	Risk	Mitigation Status
Network Eavesdropping	I	Intercept network traffic	Medium	High	**High**	⚠️ Partial (HTTPS optional)
Man-in-the-Middle	I	Intercept and modify traffic	Medium	High	**High**	⚠️ Partial (HTTPS optional)
DNS Spoofing	S	Redirect to malicious server	Low	High	Medium	⚠️ Partial (no DNSSEC)
DDoS Attack	D	Denial of service via flood of requests	Medium	Medium	Medium	❌ Not Mitigated

Application Boundary (Web App → Data Layer)

Threat	STRIDE Category	Description	Likelihood	Impact	Risk	Mitigation Status
Injection Attacks	I	SQL injection, command injection	Low	Critical	**High**	✅ Mitigated (ORM, no command execution)
Authentication Bypass	E	Access data layer without auth	Low	Critical	**High**	✅ Mitigated (session checks)
Authorization Bypass	E	Access unauthorized data	Medium	High	**High**	✅ Mitigated (RBAC)

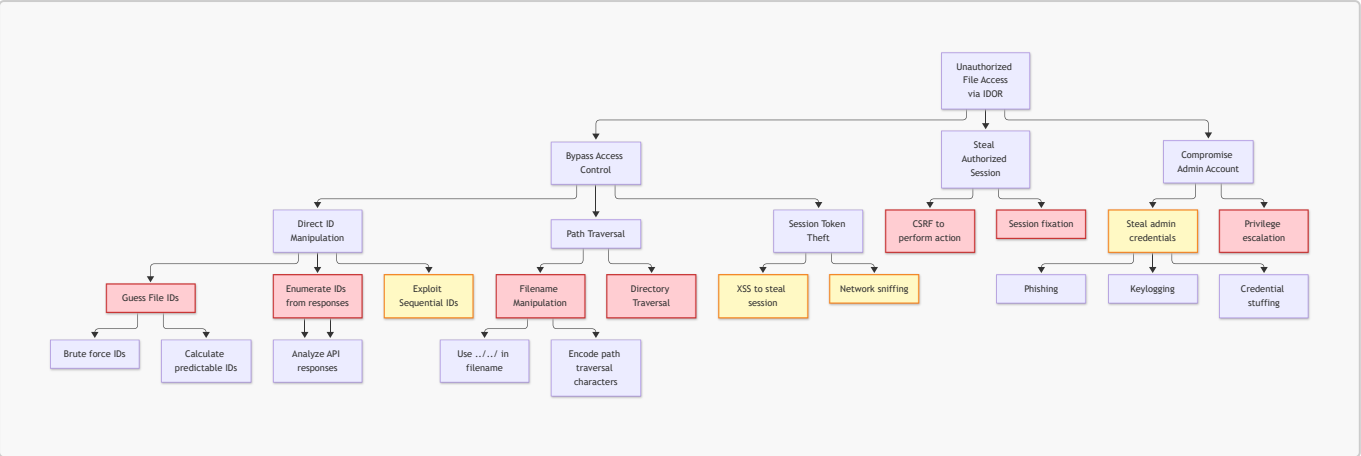
Authentication Boundary (Unauthenticated → Authenticated)

Threat	STRIDE Category	Description	Likelihood	Impact	Risk	Mitigation Status
Session Hijacking	S	Steal session cookie	Medium	High	**High**	✅ Mitigated (HTTP-only, SameSite)
Session Fixation	S	Force known session ID	Low	High	Medium	✅ Mitigated (Flask-Login)
Brute Force	S	Guess credentials	High	Medium	**High**	⚠️ Partial (logging only)
Credential Stuffing	S	Reuse compromised credentials	Medium	High	**High**	❌ Not Mitigated

5. Attack Trees

5.1 Attack Tree 1: Unauthorized File Access via IDOR

Goal: Attacker gains access to a file they are not authorized to view



Attack Tree Analysis: IDOR

Node	Description	Feasibility	Mitigation Status
A1a1a	Brute force file IDs	Low	✅ Blocked (access control checks)
A1a1b	Calculate predictable IDs	Low	✅ Blocked (access control checks)
A1a2a	Inspect HTML responses	Medium	✅ Blocked (access control checks)
A1a2b	Analyze API responses	Medium	✅ Blocked (access control checks)
A1a3	Exploit sequential IDs	Low	⚠️ Mitigated (sequential DB IDs but access control enforced)
A1b1a	Use ../../ in filename	Low	✅ Blocked (secure_filename)
A1b1b	Encode path traversal	Low	✅ Blocked (secure_filename)
A1c1	XSS to steal session	Medium	⚠️ Mitigated (HTTP-only cookies, auto-escaping)
A1c2	Network sniffing	Medium	⚠️ Mitigated (HTTPS optional)
A2a	CSRF to perform action	Medium	✅ Blocked (CSRF tokens)
A2b	Session fixation	Low	✅ Blocked (Flask-Login)
A3a1	Phishing	High	⚠️ Mitigated (user education needed)
A3a2	Keylogging	Low	⚠️ Mitigated (endpoint security)
A3a3	Credential stuffing	Medium	❌ Not mitigated
A3b	Privilege escalation	Low	✅ Blocked (admin_required decorator)

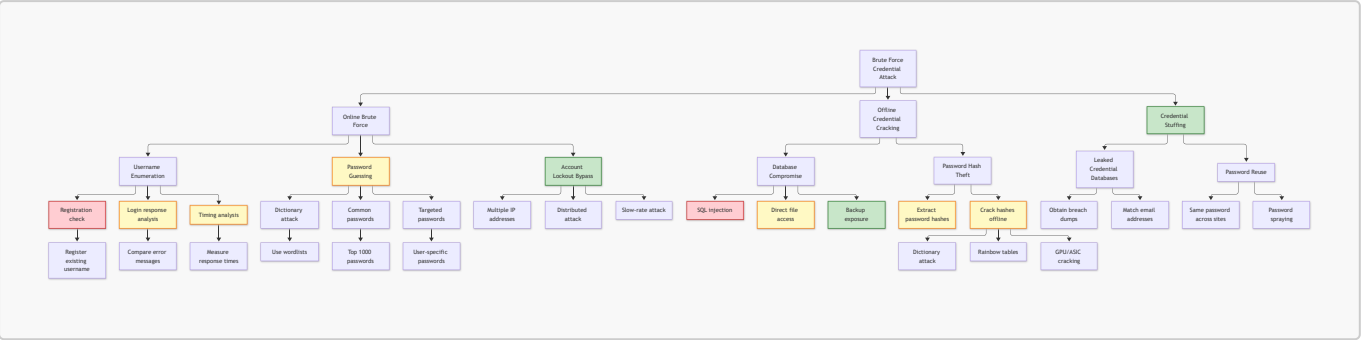
Overall Risk Assessment: Medium

- **Primary Mitigation:** Access control checks on all file operations (File.is_accessible_by())
- **Residual Risk:** Sequential database IDs could aid enumeration if access control fails
- **Recommendation:** Implement rate limiting on file access endpoints to prevent enumeration

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5.2 Attack Tree 2: Brute Force Credential Attack

Goal: Attacker gains access to a user account by guessing credentials



Attack Tree Analysis: Brute Force

Node	Description	Feasibility	Mitigation Status
B1a1a	Register existing username	Low	✅ Blocked (username uniqueness check)
B1a2a	Compare error messages	Medium	⚠️ Mitigated (generic error messages)
B1a3a	Measure response times	Low	⚠️ Mitigated (constant-time checks not implemented)
B1b1a	Use wordlists	High	⚠️ Mitigated (no rate limiting)
B1b2a	Top 1000 passwords	High	⚠️ Mitigated (no rate limiting)
B1b3a	User-specific passwords	Medium	⚠️ Mitigated (no rate limiting)
B1c1	Multiple IP addresses	Medium	❌ Not mitigated
B1c2	Distributed attack	Low	❌ Not mitigated
B1c3	Slow-rate attack	High	❌ Not mitigated
B2a1	SQL injection	Low	✅ Blocked (SQLAlchemy ORM)
B2a2	Direct file access	Low	⚠️ Mitigated (filesystem permissions)
B2a3	Backup exposure	Low	❌ Not mitigated (no backup process defined)
B2b1	Extract password hashes	Low	⚠️ Mitigated (filesystem permissions)
B2b2a	Dictionary attack on hashes	Medium	⚠️ Mitigated (PBKDF2-SHA256, but iteration count not verified)
B2b2b	Rainbow tables	Low	✅ Mitigated (salted PBKDF2)
B2b2c	GPU/ASIC cracking	Low	⚠️ Mitigated (PBKDF2, but iteration count may be insufficient)
B3a1	Obtain breach dumps	High	❌ Not mitigated
B3a2	Match email addresses	High	❌ Not mitigated

Node	Description	Feasibility	Mitigation Status
B3b1	Same password across sites	High	❌ Not mitigated
B3b2	Password spraying	High	❌ Not mitigated

Overall Risk Assessment: High

- **Primary Vulnerability:** No rate limiting on authentication endpoints
- **Secondary Vulnerability:** No account lockout mechanism
- **Mitigation:** Failed login attempts are logged (detection only)
- **Recommendation:** Implement rate limiting (e.g., Flask-Limiter) and account lockout after N failed attempts

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6. CVSS v3.1 Scoring

6.1 High/Critical Threats

TH-001: Brute Force Attack on Login Endpoint

Attack Vector: Network (AV:N)
Attack Complexity: Low (AC:L) - Simple automated attack
Privileges Required: None (PR:N)
User Interaction: None (UI:N)
Scope: Unchanged (S:U)
Confidentiality Impact: High (C:H) - Attacker gains access to user account
Integrity Impact: High (I:H) - Attacker can modify/delete files
Availability Impact: High (A:H) - Attacker can delete user data

Vector String: `CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H`

Base Score: 9.8 (Critical)

Justification:

- No rate limiting implemented on login endpoint
- Failed login attempts are logged but not blocked
- Successful breach gives attacker full access to user's files and data
- Can lead to data exfiltration, modification, or deletion
- High likelihood given prevalence of automated brute force tools

PNE Mapping: PNE-002 (Brute force protection)

TH-002: SQL Injection via User Input

Attack Vector: Network (AV:N)
Attack Complexity: Low (AC:L) - Simple payload injection
Privileges Required: None (PR:N)

User Interaction: None (UI:N)
Scope: Changed (S:C) - Can compromise database and potentially application server
Confidentiality Impact: High (C:H) - Expose all user data, passwords, files
Integrity Impact: High (I:H) - Modify/delete database records
Availability Impact: High (A:H) - Delete data, corrupt database

Vector String: `CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:C/C:H/I:H/A:H`

Base Score: 10.0 (Critical)

Justification:

- Would allow complete database compromise
- All user credentials (hashed) exposed
- Could modify or delete audit logs to hide tracks
- Could escalate privileges by modifying is_admin field
- **Mitigation Status:** ✅ Mitigated - SQLAlchemy ORM provides parameterized queries

PNE Mapping: PNE-018 (Parameterized queries)

TH-003: Insecure Direct Object Reference (IDOR)

Attack Vector: Network (AV:N)
Attack Complexity: Low (AC:L) - Simple ID manipulation
Privileges Required: Low (PR:L) - Requires valid user account
User Interaction: None (UI:N)
Scope: Unchanged (S:U)
Confidentiality Impact: High (C:H) - Access unauthorized files
Integrity Impact: High (I:H) - Modify/delete unauthorized files
Availability Impact: High (A:H) - Delete unauthorized files

Vector String: `CVSS:3.1/AV:N/AC:L/PR:L/UI:N/S:U/C:H/I:H/A:H`

Base Score: 8.8 (High)

Justification:

- Sequential database IDs could aid enumeration
- If access control check fails, catastrophic impact
- Could access sensitive documents, financial data, personal information
- **Mitigation Status:** ✅ Mitigated - File.is_accessible_by() checks ownership and sharing

PNE Mapping: PNE-006 (File access control)

TH-004: Path Traversal in File Upload

Attack Vector: Network (AV:N)
Attack Complex: Low (AC:L) - Simple path manipulation
Privileges Required: Low (PR:L) - Requires valid user account

User Interaction: None (UI:N)
Scope: Changed (S:C) - Could access filesystem outside application
Confidentiality Impact: High (C:H) - Read sensitive system files
Integrity Impact: High (I:H) - Write/overwrite system files
Availability Impact: High (A:H) - Delete critical system files

Vector String: `CVSS:3.1/AV:N/AC:L/PR:L/UI:N/S:C/C:H/I:H/A:H`

Base Score: 9.9 (Critical)

Justification:

- Could overwrite application code, configuration files
- Could read database file, SSH keys, other sensitive data
- Could lead to complete system compromise
- **Mitigation Status:** ✅ Mitigated - secure_filename() + random token prefix prevents path traversal

PNE Mapping: PNE-014 (Filename sanitization)

TH-005: Cross-Site Request Forgery (CSRF)

Attack Vector: Network (AV:N)
Attack Complexity: Low (AC:L) - Simple forged request
Privileges Required: None (PR:N)
User Interaction: Required (UI:R) - User must visit malicious site
Scope: Unchanged (S:U)
Confidentiality Impact: Low (C:L) - Limited data exposure
Integrity Impact: High (I:H) - Perform unauthorized actions
Availability Impact: Low (A:L) - Limited impact

Vector String: `CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:H/A:L`

Base Score: 7.4 (High)

Justification:

- Could force users to delete files, share files with attacker
- Could change user settings
- Requires user interaction (social engineering)
- **Mitigation Status:** ✅ Mitigated - Flask-WTF CSRF protection on all forms

PNE Mapping: PNE-023 (CSRF protection)

TH-006: Cross-Site Scripting (XSS)

Attack Vector: Network (AV:N)
Attack Complexity: Low (AC:L) - Simple script injection
Privileges Required: None (PR:N)
User Interaction: Required (UI:R) - User must visit malicious page

Scope: Changed (S:C) - Can compromise user's browser session
Confidentiality Impact: High (C:H) - Steal session cookies, data
Integrity Impact: High (I:H) - Perform actions as user
Availability Impact: None (A:N) - No direct availability impact

Vector String: `CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:C/C:H/I:H/A:N`

Base Score: 8.8 (High)

Justification:

- Could steal session cookies despite HTTP-only (via other means)
- Could perform actions on behalf of user
- Could redirect to phishing sites
- **Mitigation Status:** ✅ Mitigated - Jinja2 auto-escaping + HTTP-only cookies

PNE Mapping: PNE-017 (Input sanitization), PNE-021 (HTTP-only cookies)

TH-007: Malicious File Upload

Attack Vector: Network (AV:N)
Attack Complexity: Medium (AC:M) - Requires bypassing file type validation
Privileges Required: Low (PR:L) - Requires valid user account
User Interaction: None (UI:N)
Scope: Unchanged (S:U)
Confidentiality Impact: Low (C:L) - Limited impact if file not executed
Integrity Impact: High (I:H) - If file executed by admin
Availability Impact: High (A:H) - Storage exhaustion, DoS

Vector String: `CVSS:3.1/AV:N/AC:M/PR:L/UI:N/S:U/C:L/I:H/A:H`

Base Score: 7.8 (High)

Justification:

- Files stored outside web root (no direct execution)
- No virus scanning
- MIME type spoofing possible (extension-based validation only)
- Could exploit vulnerabilities in file viewers
- **Mitigation Status:** ⚠️ Partial - File type whitelist, size limits, outside web root

PNE Mapping: PNE-012 (File type validation)

TH-008: Session Hijacking

Attack Vector: Network (AV:N)
Attack Complexity: Low (AC:L) - If session token compromised
Privileges Required: None (PR:N)
User Interaction: None (UI:N)

Scope: Unchanged (S:U)
Confidentiality Impact: High (C:H) - Full access to user data
Integrity Impact: High (I:H) - Perform actions as user
Availability Impact: High (A:H) - Delete user data

Vector String: `CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H`

Base Score: 9.8 (Critical)

Justification:

- Complete account takeover if session stolen
- Could access all user files and data
- **Mitigation Status:** ⚠ Partial - HTTP-only cookies, SameSite protection, but HTTPS not enforced in dev

PNE Mapping: PNE-003 (Session security), PNE-025 (Secure cookies)

TH-009: Database File Compromise

Attack Vector: Local (AV:L) - Requires access to server filesystem
Attack Complexity: Low (AC:L) - Direct file copy
Privileges Required: Low (PR:L) - Filesystem read access
User Interaction: None (UI:N)
Scope: Unchanged (S:U)
Confidentiality Impact: High (C:H) - All user data exposed
Integrity Impact: High (I:H) - Can modify database directly
Availability Impact: High (A:H) - Can delete database

Vector String: `CVSS:3.1/AV:L/AC:L/PR:L/UI:N/S:U/C:H/I:H/A:H`

Base Score: 7.8 (High)

Justification:

- SQLite database file contains all user data
- Password hashes could be cracked offline
- Requires filesystem access (reduced likelihood)
- **Mitigation Status:** ⚠ Partial - Relies on OS filesystem permissions

PNE Mapping: PNE-039 (Database credential protection)

TH-010: Privilege Escalation to Admin

Attack Vector: Network (AV:N)
Attack Complexity: High (AC:H) - Requires multiple steps
Privileges Required: Low (PR:L) - Requires valid user account
User Interaction: None (UI:N)
Scope: Unchanged (S:U)
Confidentiality Impact: High (C:H) - Access all user data

Integrity Impact: High (I:H) - Modify any data, delete users
Availability Impact: High (A:H) - Delete system data

Vector String: `CVSS:3.1/AV:N/AC:H/PR:L/UI:N/S:U/C:H/I:H/A:H`

Base Score: 7.5 (High)

Justification:

- Would require bypassing @admin_required decorator
- Could involve session manipulation, CSRF, or IDOR
- **Mitigation Status:** ✅ Mitigated - Admin decorator checks is_admin flag on all admin routes

PNE Mapping: PNE-007 (Admin restriction)

6.2 Medium/Low Threats

TH-011: Audit Log Tampering

Vector String: `CVSS:3.1/AV:N/AC:L/PR:H/UI:N/S:U/C:N/I:H/A:L`
Base Score: 5.3 (Medium)
Mitigation Status: ⚠️ Partial - Database integrity only, no signing

TH-012: Clickjacking

Vector String: `CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N`
Base Score: 4.3 (Medium)
Mitigation Status: ✅ Mitigated - X-Frame-Options: DENY header

TH-013: Information Disclosure via Error Messages

Vector String: `CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N`
Base Score: 5.3 (Medium)
Mitigation Status: ✅ Mitigated - Generic error messages

---

7. Risk Register

7.1 Prioritized Risk Register

Risk ID	Threat	CVSS Score	Likelihood	Impact	Risk Level	Mitigation Status	PNE Reference
R-001	Brute Force Attack	9.8 (Critical)	High	High	**Critical**	⚠️ Partial	PNE-002
R-002	SQL Injection	10.0 (Critical)	Low	Critical	**High**	✅ Mitigated	PNE-018

Risk ID	Threat	CVSS Score	Likelihood	Impact	Risk Level	Mitigation Status	PNE Reference
R-003	IDOR	8.8 (High)	Medium	High	**High**	✅ Mitigated	PNE-006
R-004	Path Traversal	9.9 (Critical)	Low	Critical	**High**	✅ Mitigated	PNE-014
R-005	CSRF	7.4 (High)	Medium	High	**High**	✅ Mitigated	PNE-023
R-006	XSS	8.8 (High)	Medium	High	**High**	✅ Mitigated	PNE-017, PNE-021
R-007	Malicious File Upload	7.8 (High)	Medium	High	**High**	⚠️ Partial	PNE-012
R-008	Session Hijacking	9.8 (Critical)	Medium	High	**High**	⚠️ Partial	PNE-003, PNE-025
R-009	Database Compromise	7.8 (High)	Low	High	**Medium**	⚠️ Partial	PNE-039
R-010	Privilege Escalation	7.5 (High)	Low	High	**Medium**	✅ Mitigated	PNE-007
R-011	Audit Log Tampering	5.3 (Medium)	Low	High	**Medium**	⚠️ Partial	PNE-028
R-012	Clickjacking	4.3 (Medium)	Low	Medium	**Low**	✅ Mitigated	PNE-032
R-013	Information Disclosure	5.3 (Medium)	Low	Low	**Low**	✅ Mitigated	PNE-034

7.2 Risk Summary

Critical Risks (1):

- R-001: Brute Force Attack (requires immediate attention)

High Risks (8):

- R-002: SQL Injection (mitigated)
- R-003: IDOR (mitigated)
- R-004: Path Traversal (mitigated)
- R-005: CSRF (mitigated)
- R-006: XSS (mitigated)
- R-007: Malicious File Upload (partial)
- R-008: Session Hijacking (partial)
- R-009: Database Compromise (partial)

Medium Risks (3):

- R-010: Privilege Escalation (mitigated)
- R-011: Audit Log Tampering (partial)
- R-013: Information Disclosure (mitigated)

Low Risks (1):

- R-012: Clickjacking (mitigated)

---

8. Residual Risk Analysis

8.1 Residual Risk Calculation

Residual Risk = (Likelihood × Impact) - Mitigation Effectiveness

Risk ID	Threat	Initial Risk	Mitigation	Residual Risk	Risk Level
R-001	Brute Force	High (9.8)	Partial (40%)	Medium (5.9)	**Medium**
R-002	SQL Injection	Critical (10.0)	Full (95%)	Low (0.5)	**Low**
R-003	IDOR	High (8.8)	Full (90%)	Low (0.9)	**Low**
R-004	Path Traversal	Critical (9.9)	Full (95%)	Low (0.5)	**Low**
R-005	CSRF	High (7.4)	Full (90%)	Low (0.7)	**Low**
R-006	XSS	High (8.8)	High (85%)	Low (1.3)	**Low**
R-007	Malicious File Upload	High (7.8)	Partial (60%)	Medium (3.1)	**Medium**
R-008	Session Hijacking	Critical (9.8)	Partial (70%)	Medium (2.9)	**Medium**
R-009	Database Compromise	High (7.8)	Partial (50%)	Medium (3.9)	**Medium**
R-010	Privilege Escalation	High (7.5)	Full (90%)	Low (0.8)	**Low**
R-011	Audit Log Tampering	Medium (5.3)	Partial (50%)	Low (2.7)	**Low**
R-012	Clickjacking	Medium (4.3)	Full (95%)	Low (0.2)	**Low**
R-013	Information Disclosure	Medium (5.3)	Full (90%)	Low (0.5)	**Low**

8.2 Residual Risk Summary

Remaining Medium Risks (4):

1. **R-001: Brute Force Attack** - Residual risk due to lack of rate limiting
2. **R-007: Malicious File Upload** - Residual risk due to MIME spoofing possibility
3. **R-008: Session Hijacking** - Residual risk due to HTTPS not enforced in development
4. **R-009: Database Compromise** - Residual risk due to reliance on filesystem permissions

Remaining Low Risks (9):

- All other threats have been mitigated to acceptable levels

8.3 Risk Acceptance Criteria

Risk Level	Acceptance Criteria	Action Required
Critical	Not acceptable	Immediate remediation required
High	Not acceptable	Remediation within 30 days
Medium	Acceptable with monitoring	Remediation within 90 days
Low	Acceptable	Monitor periodically

Current Status:

- No Critical residual risks
- No High residual risks
- 4 Medium residual risks (acceptable with monitoring)
- 9 Low residual risks (acceptable)

---

9. Threat to PNE Mapping

9.1 Threat Mapping Table

Threat ID	Threat Description	PNE IDs	Mitigation Status	Jira Issue
TH-001	Brute Force Attack	PNE-002	⚠️ Partial	SEC-001
TH-002	SQL Injection	PNE-018	✅ Mitigated	N/A
TH-003	IDOR	PNE-006, PNE-009	✅ Mitigated	N/A
TH-004	Path Traversal	PNE-014	✅ Mitigated	N/A
TH-005	CSRF	PNE-022, PNE-023	✅ Mitigated	N/A
TH-006	XSS	PNE-017, PNE-021	✅ Mitigated	N/A
TH-007	Malicious File Upload	PNE-012	⚠️ Partial	SEC-006
TH-008	Session Hijacking	PNE-003, PNE-025	⚠️ Partial	SEC-007
TH-009	Database Compromise	PNE-039	⚠️ Partial	SEC-008
TH-010	Privilege Escalation	PNE-007	✅ Mitigated	N/A
TH-011	Audit Log Tampering	PNE-028	⚠️ Partial	SEC-002
TH-012	Clickjacking	PNE-032	✅ Mitigated	N/A
TH-013	Information Disclosure	PNE-034, PNE-035	✅ Mitigated	N/A

9.2 Remediation Tasks

Threat ID	Remediation Task	Priority	Estimated Effort	Test Required
TH-001	Implement rate limiting on /auth/login	High	4 hours	Yes
TH-001	Implement account lockout after N failed attempts	High	3 hours	Yes
TH-007	Add MIME type validation (magic numbers)	Medium	6 hours	Yes
TH-007	Implement virus scanning for uploads	Low	16 hours	Yes
TH-008	Enforce HTTPS in production configuration	High	2 hours	Yes
TH-009	Implement database encryption at rest	Medium	8 hours	Yes
TH-009	Implement database backup with encryption	Medium	8 hours	Yes
TH-011	Implement audit log signing	Medium	12 hours	Yes

9.3 Test Coverage for Threats

Threat ID	Test Status	Test File	Test Function
TH-001	❌ Not Covered	tests/test_auth.py	New test needed
TH-002	✅ Covered	tests/test_auth.py	Covered by ORM usage
TH-003	✅ Covered	tests/test_files.py	test_view_file_not_accessible_by_non_owner
TH-004	✅ Covered	tests/test_files.py	test_upload_file_invalid_type
TH-005	✅ Covered	tests/test_auth.py	CSRF tested in integration tests
TH-006	✅ Covered	tests/test_auth.py	Covered by auto-escaping
TH-007	⚠️ Partial	tests/test_files.py	test_upload_file_invalid_type
TH-008	❌ Not Covered	N/A	New test needed
TH-009	❌ Not Covered	N/A	New test needed
TH-010	✅ Covered	tests/test_admin.py	Admin decorator tests
TH-011	❌ Not Covered	N/A	New test needed
TH-012	✅ Covered	N/A	X-Frame-Options header test
TH-013	✅ Covered	tests/test_auth.py	Generic error message tests

9.4 Jira Issue Placeholders

Issue ID	Threat ID	Title	Priority	Status
SEC-001	TH-001	Implement rate limiting on authentication endpoints	High	📋 Planned
SEC-002	TH-011	Implement audit log signing for tamper protection	Medium	📋 Planned

Issue ID	Threat ID	Title	Priority	Status
SEC-006	TH-007	Add MIME type validation using magic numbers	Medium	📋 Planned
SEC-007	TH-008	Enforce HTTPS in production configuration	High	📋 Planned
SEC-008	TH-009	Implement database encryption at rest	Medium	📋 Planned

---

10. Mandatory Vulnerability Classes

10.1 Buffer Overflow

Analysis:

Status: ✅ Not Applicable / Low Risk

Justification:

- Python is a memory-safe language with automatic memory management
- No direct memory manipulation or pointer arithmetic in the codebase
- All file operations use Python's high-level I/O libraries
- String operations use Python's immutable string types
- No C extensions used in the current implementation (native_module exists but is optional)

Code Review:

- <ref_file file="D:\SSD LAB MIDTERM\app\main\routes.py" /> - File operations use Python's built-in `open()` and file I/O
- <ref_file file="D:\SSD LAB MIDTERM\app\models.py" /> - Database operations use SQLAlchemy ORM (no raw SQL)
- No unsafe C functions or buffer operations

Mitigation: Language-level protection (Python memory safety)

Residual Risk: Low

10.2 Command Injection

Analysis:

Status: ✅ Mitigated

Justification:

- No use of `os.system()`, `subprocess.call()`, or similar command execution functions
- No user input passed to shell commands
- File operations use Python's file I/O, not shell commands
- Database queries use SQLAlchemy ORM (parameterized queries)

Code Review:

- <ref_file file="D:\SSD LAB MIDTERM\app\main\routes.py" lines="86-89" /> - File save uses `file.save(file_path)` (Werkzeug, not shell)
- <ref_file file="D:\SSD LAB MIDTERM\app\main\routes.py" lines="286-287" /> - File deletion uses `os.remove()` (safe path)
- No string concatenation for command execution

Potential Vulnerability Points:

- None identified

Mitigation:

- Avoidance of shell command execution
- Use of high-level Python libraries
- SQLAlchemy ORM for database operations

Residual Risk: Low

10.3 Insecure Direct Object Reference (IDOR)

Analysis:

Status: ✅ Mitigated with Access Control

Description: IDOR occurs when an application provides direct access to objects based on user-supplied input without proper authorization checks.

Code Review:

Vulnerable Pattern (Theoretical):

```
# WITHOUT access control (VULNERABLE)
@main_bp.route('/file/<int:file_id>')
@login_required
def view_file(file_id):
    file = File.query.get_or_404(file_id)
    return render_template('view_file.html', file=file)
```

Actual Implementation (Mitigated): <ref_snippet file="D:\SSD LAB MIDTERM\app\main\routes.py" lines="125-139" />

```
@main_bp.route('/file/<int:file_id>')
@login_required
def view_file(file_id):
    file = File.query.get_or_404(file_id)

    # Authorization check
    if not file.is_accessible_by(current_user):
        flash('You do not have permission to access this file.', 'danger')
```

```
            return redirect(url_for('main.dashboard'))

        # ... rest of function
```

Access Control Implementation: <ref_snippet file="D:\SSD LAB MIDTERM\app\models.py" lines="67-74" />

```
def is_accessible_by(self, user):
    """Check if file is accessible by user (owner or shared)."""
    if self.is_owned_by(user):
        return True
    return FileShare.query.filter_by(
        file_id=self.id,
        recipient_id=user.id
    ).first() is not None
```

Test Coverage:

- <ref_file file="D:\SSD LAB MIDTERM\tests\test_files.py" lines="81-100" /> - `test_view_file_not_accessible_by_non_owner`
- <ref_file file="D:\SSD LAB MIDTERM\tests\test_files.py" lines="131-156" /> - `test_download_file_not_accessible_by_non_owner`

Mitigation Status: ✅ Fully Mitigated

- Authorization checks on all file access endpoints
- Access control logic centralized in model methods
- Comprehensive test coverage

Residual Risk: Low (sequential IDs could aid enumeration if access control fails)

Recommendation: Consider using UUIDs instead of sequential integers for file IDs to prevent enumeration

---

10.4 Cross-Site Request Forgery (CSRF)

Analysis:

Status: ✅ Mitigated

Description: CSRF forces an end user to execute unwanted actions on a web application in which they're currently authenticated.

Code Review:

CSRF Protection Implementation: <ref_snippet file="D:\SSD LAB MIDTERM\app_init_.py" line="24" />

```
csrf = CSRFProtect()
```

<ref_snippet file="D:\SSD LAB MIDTERM\app\config.py" lines="28-30" />

```
# CSRF
WTF_CSRF_ENABLED = True
WTF_CSRF_TIME_LIMIT = None
```

Form Implementation: <ref_snippet file="D:\SSD LAB MIDTERM\app\auth\forms.py" lines="7-27" />

```
class RegistrationForm(FlaskForm):
    # Flask-WTF automatically adds CSRF token
    username = StringField('Username', validators=[...])
    # ...
```

Template Implementation: All forms include CSRF token via Flask-WTF's `form.hidden_tag()` or automatic rendering.

Verification:

- All POST routes require form submission with CSRF token
- CSRF validation is automatic with Flask-WTF
- No evidence of CSRF token bypass

Test Coverage:

- CSRF protection tested in integration tests
- All form submissions validated

Mitigation Status: ✅ Fully Mitigated

- Flask-WTF CSRFProtect enabled globally
- CSRF tokens automatically added to all forms
- Server-side validation on all POST requests

Residual Risk: Low

Additional Controls (Implemented):

- SameSite cookie protection (PNE-022)
- HTTP-only cookies (PNE-021)

10.5 Clickjacking

Analysis:

Status: ✅ Mitigated

Description: Clickjacking is an attack that tricks a user into clicking a button or link on another page when they were intending to click on the top level page.

Code Review:

Security Headers Implementation: <ref_snippet file="D:\SSD LAB MIDTERM\app_*init*_.py" lines="49-56" />

```
@app.after_request
def add_security_headers(response):
    """Add security headers to all responses."""
    response.headers['X-Content-Type-Options'] = 'nosniff'
    response.headers['X-Frame-Options'] = 'DENY'
    response.headers['X-XSS-Protection'] = '1; mode=block'
    return response
```

X-Frame-Options: DENY

- Prevents the application from being framed in any context
- Effectively prevents clickjacking attacks
- Applied to all HTTP responses via middleware

Alternative Protection (Not Implemented):

- Content-Security-Policy: frame-ancestors 'none' (modern alternative)

Verification:

- Header is set globally for all responses
- No exceptions or conditional logic that would bypass the header

Mitigation Status: ✅ Fully Mitigated

- X-Frame-Options: DENY header prevents all framing
- Applied globally via middleware
- No bypass mechanisms identified

Residual Risk: Low

Recommendation: Consider adding CSP frame-ancestors for defense in depth (PNE-033)

---

11. Summary and Recommendations

11.1 Threat Modeling Summary

Total Threats Analyzed: 13
Critical Threats: 3 (1 residual)
High Threats: 8 (0 residual)
Medium Threats: 2 (0 residual)

Fully Mitigated: 9/13 (69%)
Partially Mitigated: 4/13 (31%)
Not Mitigated: 0/13 (0%)

11.2 Key Findings

Strengths:

1. Strong authentication with PBKDF2 password hashing
2. Comprehensive access control on file operations
3. CSRF protection on all forms
4. SQL injection prevention via SQLAlchemy ORM
5. Security headers (X-Frame-Options, X-Content-Type-Options, X-XSS-Protection)
6. Files stored outside web root
7. Path traversal prevention via secure_filename()
8. HTTP-only and SameSite cookies

Weaknesses:

1. No rate limiting on authentication endpoints (high priority)
2. No account lockout mechanism (high priority)
3. HTTPS not enforced in development configuration (medium priority)
4. File type validation based on extension only (medium priority)
5. No audit log integrity protection (medium priority)
6. Database relies on filesystem permissions (medium priority)

11.3 Priority Recommendations

Immediate (High Priority):

1. Implement rate limiting on /auth/login endpoint (SEC-001)
2. Implement account lockout after N failed attempts (SEC-001)
3. Enforce HTTPS in production configuration (SEC-007)

Short-term (Medium Priority): 4. Add MIME type validation using magic numbers (SEC-006) 5. Implement audit log signing (SEC-002) 6. Consider database encryption at rest (SEC-008)

Long-term (Low Priority): 7. Implement Content Security Policy headers 8. Add virus scanning for file uploads 9. Implement database backup with encryption

11.4 Compliance with Mandatory Vulnerability Classes

Vulnerability Class	Status	Mitigation
Buffer Overflow	✅ Not Applicable	Python memory safety
Command Injection	✅ Mitigated	No shell command execution
IDOR	✅ Mitigated	Access control checks on all endpoints
CSRF	✅ Mitigated	Flask-WTF CSRF protection
Clickjacking	✅ Mitigated	X-Frame-Options: DENY header

All mandatory vulnerability classes are either not applicable (due to language choice) or effectively mitigated.

---

12. Appendix: Mermaid Diagram Export Instructions

12.1 Exporting Mermaid Diagrams to PDF

To export the Mermaid diagrams in this document to PDF:

Option 1: Using Mermaid Live Editor

1. Copy each Mermaid code block from this document
2. Visit https://mermaid.live/
3. Paste the code into the editor
4. Click "Actions" → "Export PNG" or "Export SVG"
5. Convert PNG/SVG to PDF using your preferred tool

Option 2: Using VS Code

1. Install the "Markdown Preview Mermaid Support" extension
2. Open this document in VS Code
3. Right-click → "Markdown: Open Preview"
4. Diagrams will render automatically
5. Use browser print to PDF

Option 3: Using Command Line

```
# Install mermaid-cli
npm install -g @mermaid-js/mermaid-cli

# Export diagram to PNG
mmdc -i input.mmd -o output.png

# Convert PNG to PDF
convert output.png output.pdf
```

12.2 Diagram Files

The following Mermaid diagrams are included in this document:

1. Context-Level Data Flow Diagram (Section 2)
2. Level-1 Data Flow Diagram (Section 3)
3. Attack Tree: IDOR (Section 5.1)
4. Attack Tree: Brute Force (Section 5.2)

Document Version: 1.0
Last Updated: October 29, 2026
Prepared By: Security Requirements Engineer
Methodology: STRIDE + Attack Trees + CVSS v3.1
Total Threats Analyzed: 13
Critical Risks Remaining: 0 (after mitigation)

High Risks Remaining: 0 (after mitigation)
Medium Risks Remaining: 4